

Nikhil Obuleni

Tel: (571) 478-2007 | Email: nikhil.obuleni@gwu.edu | [LinkedIn](#) | [Github](#) | [Portfolio](#) | Washington, DC

EDUCATION

George Washington University

Master of Science in Data Science | GPA: 3.75

Jan 2025 – Dec 2026

Washington, DC

- **Coursework:** Machine Learning, Deep Learning, Statistical Modeling, Data Mining, Natural Language Processing, Data Visualization, Database Management, GIS
- **Awards:** Global Leaders (CCAS) Fellowship \$17,100 (2025)

EXPERIENCE

Center for Global Mental Health Equity (CGMHE)

Data Scientist

May 2026 – Present

Washington, DC

- Built NLP evaluation pipelines (OpenAI API, Hugging Face Transformers) to score ChatGPT counselor competency responses against multilingual clinical ground truth across 6 languages on a **\$2.9M NIMH grant** for WHO and UNICEF.
- Trained an XGBoost model (**ROC-AUC 0.91, 82% accuracy, +11pp over logistic baseline**) with SHAP feature attribution for at-risk population identification across 1,000+ clinical programs in 6 countries; findings informed prioritization decisions for WHO and UNICEF leads.
- Set up Airflow and AWS (S3, EC2) pipelines ingesting **500K+ patient records** into PostgreSQL across 25+ countries; standardized ETL with Python, PySpark, SQL, and R cutting integration latency **83%** and reporting time **60%**.

Data Science for Sustainable Development

Data Scientist (Part-time)

Feb 2025 – May 2026

Washington, DC

- Developed ARIMA, LSTM, and XGBoost ensemble models on **200K+ humanitarian records** across 18 countries, achieving **78% weighted F1** on disaster impact forecasting for field teams with limited data infrastructure.
- Standardized data sharing workflows across **6 NGO partners** using Apache Airflow DAGs on BigQuery, cutting cross-organization integration time from **14 days to 48 hours** and enabling near real-time reporting for humanitarian operations.
- Performed exploratory data analysis and feature engineering on multi-country humanitarian datasets; designed evaluation metrics aligned with field deployment constraints, ensuring model outputs were interpretable and actionable for non-technical stakeholders.

Asteria Aerospace

Data Science Intern

May 2023 – Aug 2023

Bangalore, India

- Built computer vision models using YOLOv5 and Faster R-CNN for real-time drone object detection, enhancing surveillance efficiency by **30%** and improving target recognition accuracy across varied environmental conditions.
- Deployed predictive maintenance models using LSTMs and Random Forest to forecast drone failures, reducing unexpected breakdowns by **25%**; designed an anomaly detection system using Autoencoders and Isolation Forest, improving fault detection by **40%**.
- Conducted geospatial data analysis using GIS, LiDAR, and satellite imagery, improving drone navigation accuracy by **20%** and reducing collision risks by 15%; communicated model results to engineering and operations stakeholders.

TECHNICAL SKILLS

Languages: Python, R, SQL

ML & Modeling: XGBoost, LightGBM, scikit-learn, PyTorch, TensorFlow, BERT, ARIMA, LSTM, PySpark, GNN, YOLOv5, Faster R-CNN, Autoencoders, Isolation Forest, Random Forest

GenAI & LLMs: OpenAI API, Hugging Face Transformers, LangChain, RAG, Prompt Engineering, SHAP Explainability

MLOps & Cloud: AWS (S3, EC2, Lambda, SageMaker), GCP (BigQuery), Docker, MLflow, Apache Airflow, PostgreSQL, MongoDB, Git, CI/CD

Methods: A/B Testing, Hypothesis Testing, Time Series Forecasting, Statistical Modeling, Feature Engineering, Causal Inference

Visualization: Tableau, Power BI, Matplotlib, Seaborn, Plotly

PROJECTS

Mental Health AI Prediction System – [GitHub](#) / *Python, TensorFlow, NLP, scikit-learn, Flask*

2026

- Built a multi-modal NLP and ML pipeline achieving **94% accuracy** in mental health condition prediction across 3 diagnostic categories; trained on 10K+ clinical screening records with TF-IDF feature extraction and ensemble classification.
- Deployed an emotionally-aware chatbot using TensorFlow and sequence-to-sequence modeling for personalized therapeutic response generation – directly informed by clinical ground truth from GWU's mental health research program.

Clinical RAG Evaluation System – [GitHub](#) / *LangChain, Pinecone, GPT-4, FastAPI, AWS Lambda, RAGAS*

2026

- Built a RAG pipeline over **5,000+ WHO and NIMH clinical guidelines** (LangChain, Pinecone, semantic chunking), deployed on AWS Lambda via FastAPI; achieved **91% response relevance** and **38% hallucination reduction** vs. GPT-4, benchmarked via RAGAS.
- Integrated prompt injection guardrails and LLM evaluation logging via LangSmith; cut clinician response-verification time **45%** in usability testing with 3 clinical reviewers.

Netflix Recommendation & Churn Prediction / *Python, PyTorch, GNN, XGBoost, LSTM*

2026

- Created a **Graph Neural Network and Collaborative Filtering** recommendation engine on **10M+ viewer records** boosting engagement by **35%**; trained an XGBoost and LSTM/ANN churn prediction model achieving **92% accuracy**, reducing subscription churn by 20%.